

GarmentERP BD

Complete Enterprise Resource Planning & Quality Assurance System Architecture Specification

GarmentERP BD is a mission-critical, enterprise-grade ERP and Quality Management System engineered specifically around the authentic operational workflows of the Bangladesh Ready-Made Garments (RMG) manufacturing sector. It unites the complete factory journey from **Buyer Inquiries & POs** to **Costing, BOM, Procurement, Warehouse Inventory, Production (Cutting, Sewing, Finishing), Carton Ratio Packing, Comprehensive QA/QC (Fabric 4-Point, Inline DHU, Rework, CAPA, AQL 2.5)**, and **Commercial Shipment**.

31

INTEGRATED WORKFLOWS

14

ROLE-BASED PERSONAS

8

DEVELOPMENT PHASES

56 / 56

AUTOMATED SQA TESTS PASSED

5

POSTMAN SQA COLLECTIONS

100%

TRACEABLE PO PIPELINE

Executive Architecture Principle: Interconnected Single Source of Truth

GarmentERP BD is engineered as an interconnected business ecosystem. Quantities and specifications are logically coupled across departments—a quantity adjustment in a Buyer PO automatically cascades into BOM requirements, triggers Procurement shortage alerts, governs Production cut plies, enforces Finishing carton limits, gates Quality thresholds, and controls Export commercial invoices.

Architect & Author

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Target Industry

Bangladesh RMG Garment Manufacturing (Savar, Gazipur, Chattogram)

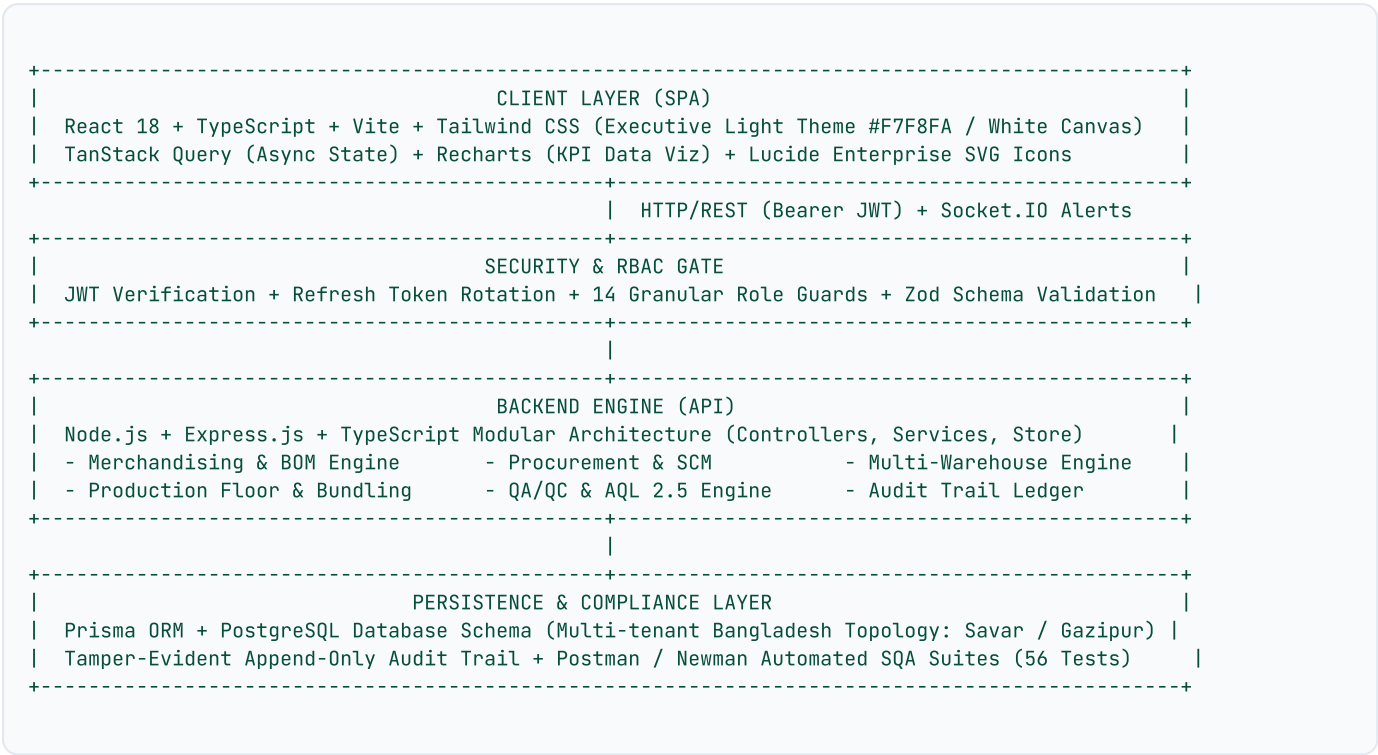
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1. Executive Summary & System Architecture

The Ready-Made Garments industry in Bangladesh represents over 84% of national exports. However, modern manufacturing plants often suffer from operational silos, where Merchandising, Commercial, Stores, Production, and Quality teams operate on disparate spreadsheets, paper tickets, and disconnected software. GarmentERP BD replaces this fragmentation with an auditable, real-time platform.

1.1 Architectural Topology



1.2 Core Technology Stack Summary

TIER	TECHNOLOGY	SPECIFICATION & ARCHITECTURAL PURPOSE
Frontend	React 18, TypeScript, Vite, Tailwind CSS	Zero-lag Single Page Application built on modern enterprise light theme design guidelines (#F7F8FA background, #FFFFFF cards, Deep Green #047857 primary accent).
State & Forms	TanStack Query, React Hook Form, Zod	Optimistic server-state caching, strict client-side validation schemas, and real-time query invalidation.
Backend API	Node.js, Express.js, TypeScript	Type-safe RESTful API architecture with strict modular domain separation, centralized middleware, and error handling.
Security	JWT, bcryptjs, RBAC Middleware	Stateless access tokens with refresh tokens, password hashing, and segregation of duties across 14 personas.
Database	PostgreSQL & Prisma ORM	Relational schema modeling Bangladesh factories, buildings, floors, lines, and warehouses.
Quality Assurance	Node Test Runner, Postman, Newman	Automated regression test suites covering positive, negative, and boundary conditions across all ERP endpoints.

1.3 UI/UX Design System Guidelines

In accordance with enterprise ERP standards, GarmentERP BD strictly avoids gimmicky dark dashboards, neon accents, or bloated admin templates. It adheres to an **Executive Light Theme**:

Design Tokens & Palette

Background Canvas	#F7F8FA (Very Light Gray)
Surfaces & Cards	#FFFFFF (Pure White)
Primary Brand	#047857 / #065F46 (Deep Green)
Secondary / Badges	#ECFDF5 (Soft Tint Green)
Alerts / Warnings	#F59E0B (Amber / Gold)
Text & Typography	#0F172A (Dark Slate / Inter)
Borders	#E2E8F0 (Subtle Slate Border)

Enterprise Principles

Collapsible Navigation	Thin right border, grouped modules
Compact KPI Cards	Large numbers with subtle percentage indicators
Data-Dense Tables	Search, filter, badges, and export actions
Order Details Tabs	Comprehensive 360° progress pipeline
Role Switcher	1-click instant persona switching for testing

2. 14 Role-Based Access Personas & Factory Topology

Strict segregation of duties is mandatory under international social compliance standards (WRAP, BSCI, SEDEX). GarmentERP BD enforces role-based access control across 14 factory roles:

2.1 The 14 Specialized Personas

ROLE PERSONA	CODE	FACTORY SCOPE & SEGREGATED OPERATIONAL RESPONSIBILITIES
Super Administrator	SUPER_ADMIN	Complete platform sovereignty, user provisioning, global permissions, and audit log administration.
Factory Administrator	FACTORY_ADMIN	Factory layout management (Savar & Gazipur units), floor and line setup, and staff roster assignments.
Executive Management	MANAGEMENT	Overall operational oversight, order profitability, costing sheet sign-offs, and shipment authorizations.
Merchandiser	MERCHANDISER	Buyer relationships, style masters, tech pack versioning, consumption BOMs, and order tracking.
Purchase Officer	PURCHASE_OFFICER	Purchase requisitions, supplier scorecards, supplier PO issuance, and raw material procurement.
Store Officer	STORE_OFFICER	Multi-warehouse stock control (Fabric, Trims, Accessories, FG), GRN entry, negative stock prevention.
Production Planner	PROD_PLANNER	Line capacity allocation, target SMV calculations, cutting schedules, and sewing plan balancing.
Production Supervisor	PROD_SUPERVISOR	Hourly sewing floor tracking (Lines 01-04), cut bundle intake, machine downtime, and output verification.
QA Manager	QA_MANAGER	Quality standards enforcement, CAPA 5-Whys verification, AQL audit approvals, and shipment quality gates.
QC Inspector	QC_INSPECTOR	4-Point fabric inspection, cutting panel QC, inline sewing defect logging (DHU), and rework inspection.
Commercial Officer	COMMERCIAL_OFFICER	Export documentation, Letters of Credit (LC), customs invoices, packing lists, and security gate passes.
HR Officer	HR_OFFICER	Employee attendance, shift rosters, overtime computation, biometric integration, and monthly payroll.
Finance Officer	FINANCE_OFFICER	Cost center allocations, supplier payables, buyer receivables, CM earnings, and order net margin analysis.
Maintenance Officer	MAINTENANCE_OFFICER	Sewing machine maintenance schedules, breakdown work orders, spare parts consumption, and downtime.

2.2 Factory Topology & Spatial Hierarchy (Module 02)

Enterprise Holding (Apex Garments Holdings Ltd.)

Plant 01: Savar Manufacturing Complex (Code: AG-SAVAR-01)

Building 01 (Main Production Block)

Floor 01: Warehouse & Fabric Store (Bins: FAB-01 to FAB-18)

Floor 02: Automated Spreading & Cutting Floor (Tables C1 - C6, 24h Relaxation Zone)

Floor 03: Sewing Department (Production Lines 01 to 04 with Juki DDL-9000C)

Floor 04: Finishing, Metal Detection & Export Packing Section

Building 02 (Commercial & Admin Center)

Plant 02: Gazipur Washing & Denim Unit (Code: AG-GAZI-02)

- └ Floor 01: Wet Processing & Sustainable Washing Department
- └ Floor 02: Finishing, AQL Final Audit Bay & Finished Goods Bonded Store

3. Commercial & Supply Chain Modules (Phases 2 & 3)

The commercial and supply chain backbone ensures that every yard of fabric and button ordered matches the exact requirements of international buyer contracts, preventing stock hoarding and negative inventory.

3.1 Commercial Workflow (Buyer PO, Tech Pack & Costing)

Buyer & Tech Pack Engine (Mod 04 & 05)

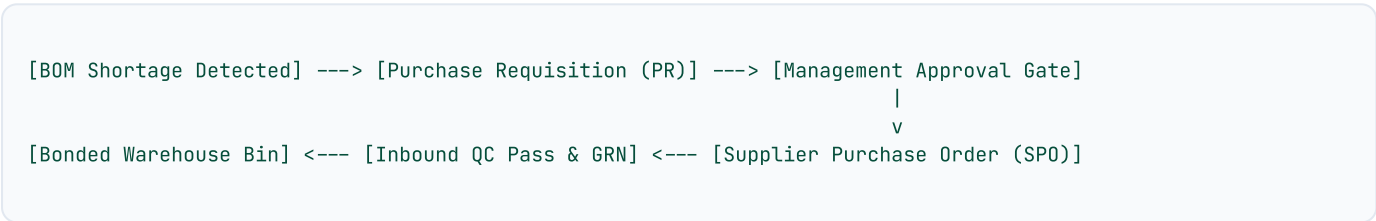
Supports global buyers (e.g., H&M, Zara, PVH, XYZ Fashion) with multi-currency (USD, EUR, GBP, BDT) and international payment terms (At-Sight LC, Usance LC). Tech packs feature strict **version control (v1, v2, v3)**, ensuring that revision changes in measurement charts or tolerances (+/- 0.5cm) notify production supervisors immediately.

BOM & Pre-Costing Engine (Mod 07 & 08)

Calculates comprehensive garment unit costs: Fabric, Trims, CM (Cost of Making), Wash, Print/Embroidery, Commercial Overhead, and Net Profit Margin. The BOM explosion multiplies item consumption (e.g. 0.85 KG single jersey/piece) by PO volume to compute gross requirements and auto-detect raw material shortages.

3.2 Supply Chain & Inventory Control (Mod 09 & 10)

Raw materials represent 60–70% of RMG order value. GarmentERP BD automates procurement from shortage detection to warehouse bin allocation:



3.3 Strict Negative Inventory Protection Engine

Core Guard: Hard Negative Stock Prevention

The system strictly disallows stock balance deductions that would result in negative quantities. If a supervisor attempts to issue 8,000 meters from a fabric lot containing only 5,000 meters, the API transaction is immediately rejected with 400 Bad Request: Insufficient Stock Available and an audit violation is logged. Over-issuance is strictly locked unless an authorized Factory Administrator enters a certified override code.

TRANSACTION TYPE	TRIGGER EVENT	SOURCE LOCATION	DESTINATION	VALIDATION RULES
GRN Receive	Vendor delivery at security gate	Supplier Inbound Truck	Quarantine Inspection Bay	Matches SPO quantity (+/- 5% allowance) & QA certificate.
QC Bin Transfer	Fabric 4-point inspection pass	Quarantine Bay	Bonded Fabric Warehouse	Roll tagged with barcode, length, and dye lot number.
Line Issue	Daily production cutting plan	Fabric Warehouse	Cutting Floor Lay Table	Deduction must be ≤ Available Unreserved Stock.
Stock Adjustment	Monthly physical inventory count	Warehouse Bin	Adjustment Ledger	Requires dual-signature from Store Officer and Finance.

4. Garments Production Management (Phase 4)

Production execution requires synchronized coordination across Cutting, Sewing, Finishing, and Carton Packing. GarmentERP BD provides end-to-end floor visibility and automated bundle routing.

4.1 Cutting Workflow & 24-Hour Fabric Relaxation Timer (Mod 12)

In knitted garment manufacturing, elastane/cotton fabric rolls undergo high mechanical tension during knitting and dyeing. Spreading rolls immediately causes post-cut shrinkage and dimensional failures.

24h Relaxation Countdown Engine

Upon receiving fabric from the warehouse, the roll enters the unrolling cradle and a **24-Hour Digital Relaxation Timer** is initiated. The system locks cut order generation against that roll until the tension-release countdown reaches zero and QC certifies dimensional stability.

Marker Efficiency & Spreading Lay

Tracks marker length, fabric width, plies count, and marker efficiency percentage (typically 84–89%). Minimizes end-loss wastage and generates automated piece yield summaries for fabric reconciliation against BOM standards.

4.2 QR / Barcode Bundle Ticketing Registry

Once cutting is executed, cut panels are bundled into standard lots (e.g., 250 pieces per bundle). Each bundle receives a tamper-evident QR code ticket encoding: Cut Order ID, Bundle #, Size, Color, and Piece Serial Range (e.g., #001 to #250). Bundles are scanned as they move across workstations for real-time WIP (Work-in-Progress) tracking.

4.3 Sewing Line Hourly Tracking (Mod 13)

Tracks production floor output hour-by-hour across lines (e.g., Lines 01 to 04 in Savar Unit) against target Standard Minute Values (SMV):

Line ID	Assigned Garment	Hourly Target	Actual Made	Rejects/Rwk	Line Efficiency
Line 01	Mens Polo Shirt	150 pcs/hr	142 pcs/hr	3 / 2 pcs	94.6% (On Target)
Line 02	Basic Crewneck	180 pcs/hr	175 pcs/hr	4 / 3 pcs	97.2% (Optimal)
Line 03	Raglan Hoodie	120 pcs/hr	102 pcs/hr	6 / 5 pcs	85.0% (Warning)
Line 04	Chino Trousar	110 pcs/hr	105 pcs/hr	2 / 1 pcs	95.4% (Optimal)

4.4 Finishing Section & Ratio Carton Packing (Mod 14 & 15)

Stitched garments move through 4 finishing stages: **Thread Trimming & Suction** → **Steam Pressing & Ironing** → **Folding & Hangtag Attachment** → **Calibrated Needle/Metal Detection**.

Packing Ratio Assortment & Over-Packing Security Guard

GarmentERP BD enforces buyer-specified carton ratio matrices (e.g., Carton #001: Size S:10, M:20, L:20, XL:10 = 60 pieces). Furthermore, the packing module strictly prevents Packed Quantity > Available Finished Quantity, guaranteeing zero phantom carton records.

5. Core QA/QC Suite — The Quality Flagship (Phase 5)

Quality assurance is the crown jewel of GarmentERP BD. It embeds international inspection methodologies across raw materials, in-process sewing, rework loops, root-cause CAPA, and pre-shipment AQL auditing.

5.1 Fabric 4-Point Inspection Engine (ASTM D5430 / Mod 16)

Every incoming fabric roll undergoes 4-point penalty inspection. Defects are assigned points based on physical defect size:

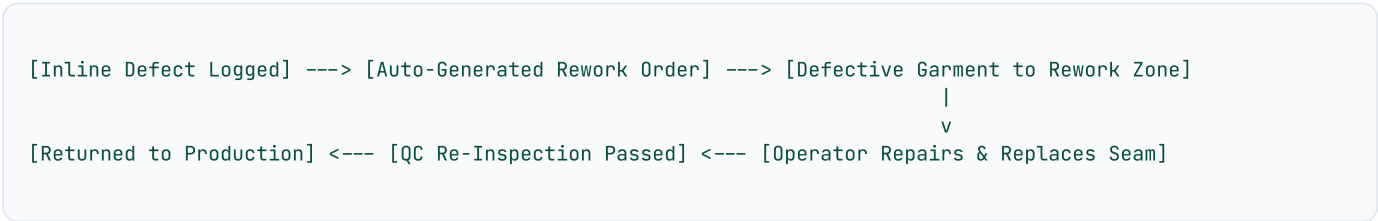
DEFECT LENGTH / SIZE RANGE	PENALTY POINTS	INDUSTRY STANDARD FORMULA & ACCEPTANCE THRESHOLD
Defects up to 3 inches	1 Point	$\text{Points per } 100 \text{ Sq. Yards} = \frac{\text{Total Penalty Points}}{\text{Inspected Yards}} \times \text{Fabric Width (Inches)}$ FIRST QUALITY PASS: ≤ 28.0 POINTS REJECTED / SECOND QUALITY: > 28.0 POINTS
Defects over 3 inches up to 6 inches	2 Points	
Defects over 6 inches up to 9 inches	3 Points	
Defects over 9 inches or any hole	4 Points	

5.2 Inline Sewing Quality & Real-Time DHU Engine (Mod 17)

QC roving inspectors record defects at operator stations in real-time. Defect categories include Broken Stitch, Skip Stitch, Open Seam, Puckering, Shading, and Oil Stains. The system automatically computes:

$$\text{DHU (Defects per Hundred Units) \%} = (\text{Total Defect Count} / \text{Total Inspected Pieces}) \times 100$$

5.3 Closed-Loop Rework Quarantine & Re-Inspection (Mod 18)



5.4 Root-Cause CAPA 5-Whys Analysis (Mod 19)

When high-severity defect patterns emerge, QA Managers launch Corrective and Preventive Actions (CAPA) utilizing the standardized **5-Whys Root-Cause Methodology**:

Case Study: Excessive Skip Stitches on Sewing Line 02

1. Why did the defect happen?

Needle thread skipped bobbin hook during placket stitch.

2. Why did it skip the hook?

Needle temperature was too high, causing thermal thread deflection.

3. Why was temperature high?

Wrong needle point geometry (Standard R used instead of SES ballpoint).

4. Why was wrong needle used?

Mechanic ran out of SES needles and fitted standard needle.

5. Root Cause (5th Why):

Preventive maintenance store had no automatic re-order point for SES needles.

Corrective Action: Fitted titanium SES ballpoint needle. | Preventive Action: Established minimum safety stock in ERP with automated purchase triggers. | Verified by: QA Manager.

5.5 ISO 2859-1 / ANSI ASQ Z1.4 Normal Level II AQL Sampling (Mod 20)

Pre-shipment final audits calculate sampling batches dynamically based on production lot sizes:

ORDER LOT SIZE RANGE	INSPECTION LEVEL	SAMPLE SIZE	CRITICAL (AQL 0)	MAJOR (AQL 2.5)	MINOR (AQL 4.0)
3,201 to 10,000 pcs (Code K)	Level II Normal	200 pieces	Ac: 0 Re: 1	Ac: 10 Re: 11	Ac: 14 Re: 15
10,001 to 35,000 pcs (Code L)	Level II Normal	315 pieces	Ac: 0 Re: 1	Ac: 14 Re: 15	Ac: 21 Re: 22

6. Shipment, Operations & SQA Automated Test Suite

The remaining roadmap covers export logistics, human resources, machinery upkeep, and the comprehensive automated testing suite guaranteeing zero-defect deployments.

6.1 Shipment & Logistics (Phase 6 — Upcoming)

Security Lock: Mandatory Quality Gate Enforced Prior to Shipment

Under international shipping protocols, GarmentERP BD strictly locks Commercial Invoicing and Gate Pass issuance if the PO's pre-shipment AQL inspection is missing or has a REJECTED status. No container can be dispatched from the factory without an authorized digital QA sign-off certificate.

Commercial Invoicing & Packing List

Generates customs-compliant export documentation with gross weight, net weight, CBM (Cubic Meter) container calculations, and Harmonized System (HS) codes.

Security Gate Pass & Dispatch Tracking

Issues tamper-proof QR gate passes for logistics container trucks dispatched to Chittagong Port or Dhaka Airport (In-Transit → Port → Delivered → Closed).

6.2 Factory Operations & HR (Phase 7 — Upcoming)

HR, Shifts & Biometric Payroll

Employee Roster	Operators, helpers, supervisors
Shift Scheduling	Day, evening, night rotations
Overtime Calculation	Compliant with BD Labor Act 2006
Biometric Sync	Automated punch-card integration

Machine Maintenance & Downtime

Machine Registry	Juki, Brother, Pegasus, Kansai
Preventive Maintenance	Daily/Weekly lube & oil schedules
Breakdown Tickets	Mechanic response time & downtime
Spare Parts Cost	Needles, loopers, belts, motors

6.3 SQA Verification: 56 Automated Tests Passing (100% Pass Rate)

TEST SUITE MODULE	TESTS RUN	PASS / FAIL	KEY AUTOMATED COVERAGE & ASSERTIONS VERIFIED
Phase 1: Auth & RBAC	10 Tests	10 PASS / 0 FAIL	JWT login, 401 invalid password rejection, RBAC 403 authorization guard, factory topology retrieval, audit trail logging.
Phase 2: Merchandising & BOM	10 Tests	10 PASS / 0 FAIL	Buyer profile creation, Tech pack v1-v3 versioning, Costing sheet profit margin calculation, PO 5,000 pcs matrix, BOM shortage auto-calculation.
Phase 3: SCM & Inventory	12 Tests	12 PASS / 0 FAIL	Supplier registration & scorecards, PR generation, SPO dispatch, GRN receipt, Strict Negative Inventory 400 rejection on over-issuance.
Phase 4: Production Floor	12 Tests	12 PASS / 0 FAIL	24h fabric relaxation timer, cut order marker efficiency %, QR bundle ticketing (250 pcs), Lines 01-04 hourly efficiency, ratio carton packing.
Phase 5: Core QA/QC Suite	12 Tests	12 PASS / 0 FAIL	Fabric 4-point calculation (2.0 pts Pass vs 36.0 pts Reject), sewing DHU logging, Rework order repair-to-re-inspection, CAPA 5-Whys, ISO AQL 2.5 sampling.
TOTAL REGRESSION VERIFICATION	56 Tests	56 PASS / 0 FAIL (100%)	Continuous integration verification executed via scripts/test-all.js with 5 Postman Collections ready for Newman execution.

6.4 Complete PO Journey 360° Traceability (Phase 8 Flagship)

P0-2026-001 (XYZ Fashion | 10,000 Pcs Polo)

- └─ Merchandising: Tech Pack v2 (Measurement Tol: +/-0.5cm) | Pre-Costing (\$6.50/pc | Margin: 18.7%)
- └─ BOM & MRP: Fabric Required 8,500 KG | Available 5,000 KG | Shortage: 3,500 KG
- └─ SCM & Procurement: PR-004 → Sourcing Approval → SPO-008 → GRN-012 Inbound QC Pass
- └─ Warehouse: Fabric Lot #FL-889 → 4-Point QC Pass (18 pts/100 sq yd) → Bin FAB-04
- └─ Production: 24h Relaxation Complete → Cut Order C-102 (Marker Eff: 88.4%) → 40 Bundles (QR)
- └─ Sewing Floor: Line 01 (142 pcs/hr | Eff: 94.6%) → Inline Defect Logged → Rework → Re-Inspected Pass
- └─ Finishing & Packing: Steam Pressed & Metal Checked → 166 Cartons Ratio Packed (S:10, M:20, L:20, XL:10)
- └─ Quality Assurance: Pre-Shipment ISO 2859-1 Level II Normal AQL 2.5 Audit (Sample: 315 pcs | PASS)
- └─ Shipment & Commercial: Commercial Invoice #CI-9941 → Gate Pass #GP-204 → Chittagong Port Dispatch